$$p_2 = \frac{4\sigma}{R_2} - \frac{q^2}{32\pi^2\varepsilon_0 R_2^4}. \quad (3)$$

$$V_1 = 4\pi R_1^3/3,\ V_2 = 4\pi R_2^3/3. \quad (4)$$

(1) - (4)

$$q = 32\pi\sqrt{\varepsilon_0\sigma R_1^3}. \quad (5)$$

평가 항목	배점
$P_{initial} = P_{surf}$	0.5
$P_{final} = P_{surf} - P_{electr}$	0.3
$P_{surf} = 4\sigma/R$	0.3
$P_{electr} = q^2/32\pi^2\varepsilon_0 R^4$	0.5
	0.5
$W_{surf} = 8\pi R^2\sigma$	0.3
$W_{electr} = q^2/8\pi\varepsilon_0 R$	0.5
$W_{gas} = (5/2)\nu RT = (5/2)PV$	0.4
	0.2
	0.5
합계	**4.0**

문제 1C (3.0점)

. S_1

S_2 , 3

S_1 S_2 .

3 $\frac{\lambda}{2} + m\lambda$.

$m = 0, \pm1, \pm2...$ . .

$R_1 + \frac{\lambda}{2} + m\lambda$ A_1 ,

$R_2 + \frac{\lambda}{2} + m\lambda$ A_2 .

3 , .

3 2 .

, S_1 S_2 3 4

. , $4I_0$.

